PROJECT PLANNING PHASE

Project Planning Template (Backlog, Sprints, Stories & Points)

Date: 19 July 2026
Team ID: TEAM-VOICE-AI-09
Project Name: Voice-Based Concept Understanding Analyser
Maximum Marks: 5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

The product backlog has been aligned to implement core technical pipelines for ingestion, transcription, semantic assessment via LLMs, and metrics dashboards.

Sprint	Epic / Requirement	Story ID	User Story / Implementation Task	Points	Priority	Team
Sprint-1	Streaming Ingestion	USN-1	As a user, I can initiate a WebRTC or native audio capture stream to transmit responses directly to the gateway engine.	2	High	Dev-01
Sprint-1	Streaming Ingestion	USN-2	As an operator, I want raw audio chunks automatically saved to an isolated archival bucket for telemetry validation.	1	High	Dev-02
Sprint-1	Transcription	USN-3	As a system, I need to pipe the ingestion stream directly through a Whisper ASR cluster to obtain raw textual formats.	3	High	Dev-01
Sprint-1	User Security	USN-4	As a system administrator, I want integration with secure OAuth paths (Gmail/Enterprise OpenID) for auth protection.	2	Medium	Dev-03
Sprint-2	Semantic Assessment	USN-5	As a teacher, I want the LLM engine to extract concepts and compare them against custom-built vector rubrics.	5	High	Dev-02
Sprint-3	Analytics UI	USN-6	As a user, I can view my granular knowledge graph discrepancy mapping on an interactive frontend dashboard.	3	High	Dev-03

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	20 Points	6 Days	10 July 2026	16 July 2026	20 Points	16 July 2026
Sprint-2	20 Points	6 Days	17 July 2026	23 July 2026	--	Pending
Sprint-3	20 Points	6 Days	24 July 2026	30 July 2026	--	Pending
Sprint-4	20 Points	6 Days	31 July 2026	06 August 2026	--	Pending

Velocity Framework & Formula

Assuming a localized iteration scope where team performance tracks against specific target timeframes, we quantify velocity. Let's calculate the team's average velocity (*AV*) per iteration unit (story points per day) given a 10-day baseline duration with a sprint capacity velocity of 20 points:

$$AV = velocity / sprint\ duration = 20 / 10 = 2$$

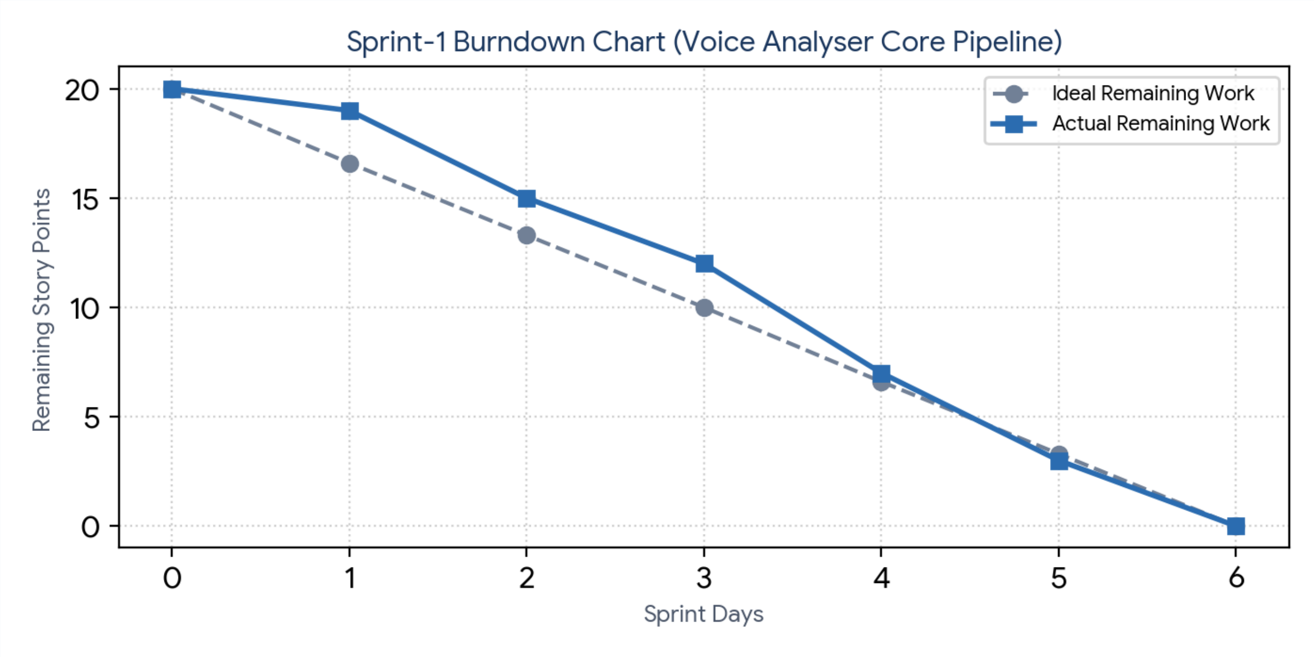

Figure 1: Sprint-1 burndown trajectory measuring daily residual story point velocity against the target baseline task group.

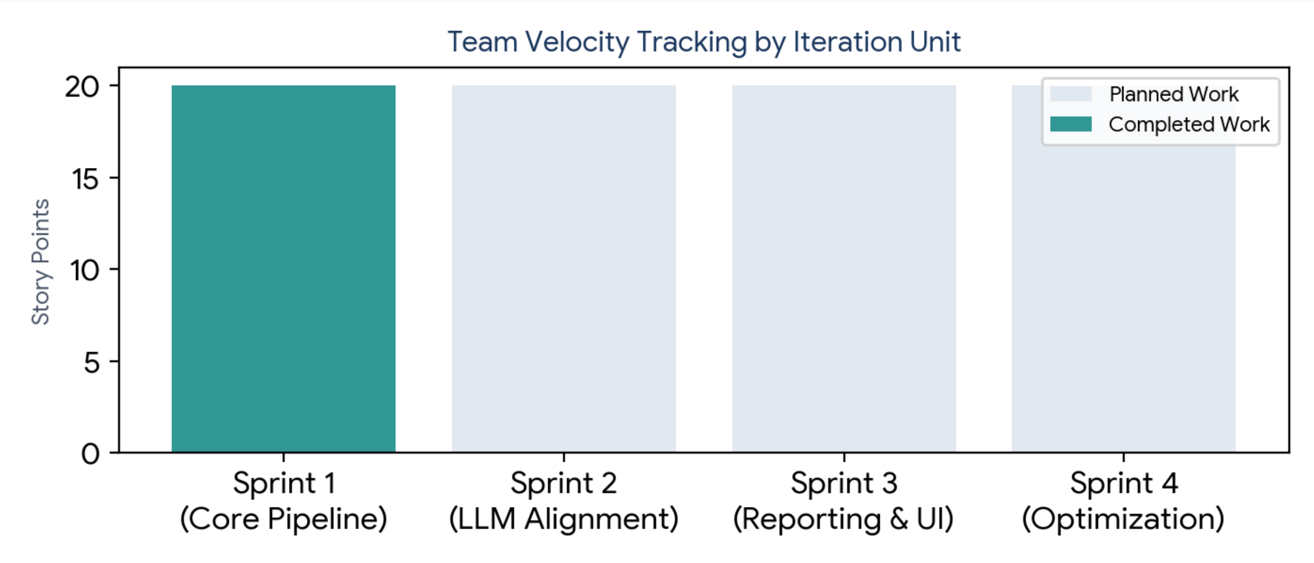

Figure 2: Comprehensive team iteration capacity map capturing actual vs planned story point processing across milestones.